# Quantum Field Theory for the Experimental Heavy-Ion Physicist

*A 14-Week Self-Study Syllabus*

Tailored for sPHENIX / STAR experimentalists building the theoretical backbone to understand QCD matter, the quark-gluon plasma, and the observables you measure every day.

### Source Texts

**[L&B]** — Lancaster & Blundell, *QFT for the Gifted Amateur*

**[P&S]** — Peskin & Schroeder, *An Introduction to Quantum Field Theory*

**[Griffiths]** — Griffiths, *Introduction to Elementary Particles* (2nd ed.)

**[Mann]** — Mann, *An Introduction to Particle Physics and the Standard Model*

**[Bartke]** — Bartke, *Introduction to Relativistic Heavy Ion Physics*

# Roadmap

The syllabus is organized into four phases that mirror how QFT knowledge feeds into heavy-ion physics. Each phase ends with a checkpoint connecting the material to experimental observables or analysis techniques you already work with.

**Phase I (Weeks 1–4)** — Foundations: Classical Fields → Canonical Quantization

> Build the scalar and Dirac field theories from the ground up. You've already done much of this—this phase consolidates and fills gaps.

**Phase II (Weeks 5–8)** — Interactions: Feynman Diagrams → QED → Cross Sections

> Learn to compute scattering amplitudes and connect them to the cross sections and decay rates you extract from data.

**Phase III (Weeks 9–12)** — QCD & Gauge Theory: Color, Confinement, Asymptotic Freedom

> The heart of the matter for heavy-ion physics. Understand QCD as a non-abelian gauge theory and why it predicts deconfinement.

**Phase IV (Weeks 13–14)** — Bridges to Heavy-Ion Physics: Symmetry Breaking, Thermal QFT, QGP

> Connect QFT to the phenomenology of the QGP: chiral symmetry breaking, the phase diagram, and signatures you look for at RHIC and the LHC.

## Estimated Workload

Plan for roughly 8–12 hours per week: ~5 hours of reading and note-taking, ~3 hours working exercises, and ~2 hours reviewing or re-deriving key results. Some weeks are lighter (marked with ★) to give breathing room around heavier material.

**How to use this guide:** The primary track uses L&B for conceptual intuition and Griffiths for computational practice. P&S is the deep reference for when you want the full formal treatment. Mann provides good Standard Model context, and Bartke anchors everything to heavy-ion observables. Sections marked *optional* are enrichment—skip on a first pass if short on time.

# Phase I: Foundations

Classical fields, canonical quantization, and the free Dirac field. You have already worked through much of this material (L&B Ch. 11, Griffiths Ch. 5–7), so this phase is about consolidating that foundation, filling in the "why," and making sure every step is airtight before moving to interactions.

---

## WEEK 1 *Lagrangian Field Theory & the Klein–Gordon Field*

**Goal:** Understand why fields (not particles) are the fundamental objects in QFT, and derive the Klein–Gordon equation from a Lagrangian.

### Reading

- **L&B** Ch. 0 (Overture), Ch. 1 (Lagrangians), Ch. 5 (Continuous systems), Ch. 6 (Klein–Gordon equation)
- **Griffiths** § 10.1–10.2 (Lagrangian formulation, relativistic field theory)
- **Mann** Ch. 4 (Conservation Laws: action principle, Noether's theorem)
- **P&S** § 2.1–2.2 (The necessity of the field viewpoint, elements of classical field theory) — *reference*

### Key Concepts to Nail Down

- Euler–Lagrange equations for fields; Hamiltonian density
- Noether's theorem: relate every continuous symmetry to a conserved current
- Klein–Gordon equation from the scalar Lagrangian; its shortcomings (negative probability density)
- Why single-particle relativistic QM fails $\rightarrow$ motivation for field theory

### Exercises

L&B Ch. 5 exercises (derive energy-momentum tensor for the scalar field). Griffiths Problem 10.1. Derive the conserved current for the complex scalar field under U(1) phase rotation.

## WEEK 2 *Canonical Quantization of the Scalar Field*

**Goal:** Promote classical fields to operators, derive the mode expansion, and understand creation/annihilation operators and normal ordering.

### Reading

- **L&B** Ch. 2–4 (SHO, occupation number, second quantization) — *review*
- **L&B** Ch. 11 (Canonical quantization of fields) — *your prior work; re-read for consolidation*
- **L&B** Ch. 12 (Quantum fields with a twist: the complex scalar field)
- **P&S** § 2.3–2.4 (The Klein–Gordon field as harmonic oscillators; the Klein–Gordon field in spacetime)

### Key Concepts to Nail Down

- Equal-time commutation relations $[\phi(x), \pi(y)] = i\delta^3(x-y)$

- Mode expansion: why each Fourier mode is an independent harmonic oscillator
- Creation/annihilation operators $a^\dagger(p)$, $a(p)$; Fock space; particle interpretation
- Normal ordering and the infinite vacuum energy (why we subtract it)
- Complex scalar field → particle vs. antiparticle (charge conjugation preview)

### Exercises

L&B Ch. 11 exercises. Verify the commutation relations for $a(p)$. Show that the Hamiltonian in terms of $a^\dagger a$ gives the particle-number interpretation. P&S Problem 2.1 (classical electromagnetism as a field theory).

## WEEK 3 *The Dirac Equation & Spinor Fields*

**Goal:** Master the Dirac equation, its solutions, gamma matrix algebra, and bilinear covariants—the language QED and QCD are written in.

### Reading

- **Griffiths** § 7.1–7.4 (Dirac equation, solutions, bilinear covariants, the photon) — *you've done this; re-derive the plane-wave solutions from scratch*
- **L&B** Ch. 36–37 (Dirac equation; how to transform a spinor)
- **Mann** Ch. 11–12 (Wave equations; gauge invariance and Dirac solutions)
- **P&S** § 3.1–3.3 (Lorentz invariance in wave equations; the Dirac equation; free-particle solutions) — *reference for the full Lorentz-group motivation*

### Key Concepts to Nail Down

- Dirac equation from "square root" of Klein–Gordon; Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$
- Bjorken–Drell convention (your preferred convention); chiral representation comparison
- Plane-wave solutions $u(p,s)$ and $v(p,s)$; completeness/spin sums
- Bilinear covariants: scalar, vector, tensor, axial vector, pseudoscalar
- Charge conjugation $\psi \to \psi^c = C\psi\blacksquare^T$

### Exercises

Griffiths Problems 7.1, 7.6, 7.8. Verify all 5 bilinear covariants transform correctly under parity. Derive the non-relativistic limit to recover the Pauli equation with the correct g=2.

## WEEK 4 *Quantizing the Dirac Field & Propagators*

**Goal:** Quantize the Dirac field with anticommutation relations, understand the spin-statistics theorem, and derive propagators for both scalar and spinor fields.

### Reading

- **L&B** Ch. 38 (The quantum Dirac field: canonical quantization, fermion propagator, Feynman rules)
- **L&B** Ch. 17 (Propagators and fields: Feynman propagator, Yukawa's force carriers)

- **P&S** § 3.5 (Quantization of the Dirac field), § 2.4 (Klein–Gordon propagator)

## Key Concepts to Nail Down

- Anticommutation relations $\{a^s(p), a^{r\dagger}(q)\} = (2\pi)^3\delta^3(p-q)\delta_{sr}$; why commutators fail for fermions
- Spin-statistics connection: Bose fields commute, Fermi fields anticommute
- The Feynman propagator as a Green's function; $i\varepsilon$ prescription and causality
- Scalar propagator $\Delta_F(x-y)$ and fermion propagator $S_F(x-y)$
- Physical meaning: propagator = amplitude for a particle to travel from y to x

**Checkpoint — connect to experiment:** The propagator is the building block of every Feynman diagram you will encounter in the rest of the syllabus. In your analysis work, cross sections and branching ratios are computed from diagrams built out of propagators and vertices. At this point, re-read Bartke Ch. 2 (§2.1–2.2 on QCD and the coupling constant) to see how these field-theory objects show up in the context of strongly interacting matter.

# Phase II: Interactions & QED

This is where QFT becomes a calculational machine. You'll learn to draw Feynman diagrams, compute amplitudes, and connect them to the cross sections and decay rates that are the bread and butter of experimental particle and heavy-ion physics.

---

## WEEK 5 *The S-Matrix, Wick's Theorem & Feynman Diagrams*

**Goal:** Derive the perturbation expansion of the S-matrix and learn to translate each term into a Feynman diagram.

### Reading

- **L&B** Ch. 18–19 (S-matrix, interaction picture, Wick's theorem; Feynman diagrams in $\phi^4$ theory)
- **Griffiths** § 6.1–6.3 (The ABC toy theory: Feynman rules, lifetimes, scattering) — *excellent for building diagram intuition*
- **Mann** Ch. 10 (A Toy Theory — Feynman rules, decay, scattering)
- **P&S** § 4.1–4.4 (Perturbation theory, Wick's theorem, Feynman diagrams) — *reference*

### Key Concepts to Nail Down

- Interaction picture (Dirac picture): operators evolve with $H_0$, states evolve with $H_{int}$
- Dyson series for the S-matrix: $S = T \exp(-i \int H_{int}\, dt)$
- Wick's theorem: convert time-ordered products into normal-ordered products + contractions
- Contractions = propagators; each contraction $\rightarrow$ an internal line in a diagram
- Symmetry factors; external lines; momentum conservation at vertices

### Exercises

L&B Ch. 19 exercises (draw all diagrams to second order in $\phi^4$). Griffiths Problem 6.1 (A lifetime), Problem 6.3 (scattering). Compute the symmetry factor for the "sunset" diagram.

## WEEK 6 *Scattering Cross Sections & Decay Rates*

**Goal:** Connect invariant amplitudes ■ to physical observables: differential cross sections $d\sigma/d\Omega$ and decay rates $\Gamma$.

### Reading

- **L&B** Ch. 20 (Scattering theory: transition matrix, invariant amplitude, cross sections)
- **Griffiths** § 6.2 (Golden Rule for decays and scattering; 2-body formulae in CM frame)
- **Mann** Ch. 9 (Scattering: lifetimes, resonances, cross sections, matrix elements, 2-body formulae)
- **P&S** § 4.5 (Cross sections and the S-matrix) — *reference*

### Key Concepts to Nail Down

- Fermi's Golden Rule in the relativistic context; phase-space integrals

- Relation: $d\sigma/d\Omega = |\blacksquare|^2 \times$ (phase space) × (flux factor)
- 2-body kinematics in CM and lab frames; Mandelstam variables s, t, u
- Breit–Wigner resonance formula — you see this every day in $\pi^0$ invariant mass fits

**Connect to your work:** The invariant mass spectra you fit for $\pi^0$ signals are literally the Breit–Wigner profile (or a Gaussian approximation of it convolved with detector resolution). The $\pi^0 \to \gamma\gamma$ branching ratio and lifetime come directly from the decay-rate calculation you're learning here.

## WEEK 7 *QED: Feynman Rules & First Calculations*

**Goal:** Derive the QED Lagrangian from gauge invariance, read off the Feynman rules, and compute electron–muon scattering.

### Reading

- **Griffiths** § 7.5–7.8 (QED Feynman rules, examples, Casimir's trick, cross sections)
- **L&B** Ch. 38.4 (Local gauge invariance → QED), Ch. 39 (Rough guide to QED: photon propagator, Feynman rules, gauge invariance)
- **Mann** Ch. 12 (Gauge invariance: the gauge principle, Maxwell–Dirac equations), Ch. 13 (QED Feynman rules, electron–muon, Bhabha, Compton scattering)
- **P&S** § 4.7–4.8, § 5.1–5.2 (QED Feynman rules, $e^+e^- \to \mu^+\mu^-$) — *reference*

### Key Concepts to Nail Down

- Local U(1) gauge invariance: $\psi \to e^{i\alpha(x)}\psi$ requires introducing $A_\mu$
- Minimal coupling: $\partial_\mu \to D_\mu = \partial_\mu + ieA_\mu$; the QED vertex = $-ie\gamma^\mu$
- Photon propagator in Feynman gauge: $-ig_{\mu\nu}/q^2$
- Casimir's trick for spin-summed/averaged $|\blacksquare|^2$; trace technology for $\gamma$-matrices
- Work through $e^-\mu^- \to e^-\mu^-$ from diagram to $d\sigma/d\Omega$

## WEEK 8 *QED Processes & Renormalization Preview ★*

**Goal:** Compute more QED processes (Compton, Bhabha, pair annihilation). Get a first look at renormalization and the running coupling.

### Reading

- **Griffiths** § 7.7–7.9 (Casimir's trick, cross sections, renormalization preview)
- **L&B** Ch. 40 (QED scattering: Rutherford, Mott, Compton, crossing symmetry)
- **L&B** Ch. 41 (Renormalization of QED: vacuum polarization, running $\alpha$, g–2)
- **Mann** Ch. 14 (Testing QED: anomalous moments, Lamb shift, running coupling)
- **L&B** Ch. 32–34 (Renormalization overview, counterterms, RG) — *skim for conceptual picture*

### Key Concepts to Nail Down

- Crossing symmetry: relate scattering, annihilation, and pair-production diagrams

- UV divergences in loop diagrams; why they appear and what they mean physically
- Renormalization in one sentence: absorb infinities into redefined (physical) mass and charge
- Running coupling $\alpha(q^2)$: the fine-structure constant is not constant!
- Vacuum polarization = the vacuum is a dielectric; screening of charge

**Checkpoint — connect to experiment:** The running of $\alpha_s$ in QCD (which you'll study next) is the mirror image of QED's running: anti-screening instead of screening, leading to asymptotic freedom. This is *the* reason the QGP exists. Understanding how renormalization works in QED makes the QCD story click.

# Phase III: QCD & Gauge Theories

This is the phase that matters most for your research. QCD is the theory that governs everything happening inside the QGP, and understanding it as a non-abelian gauge theory is essential for interpreting jet quenching, collective flow, strangeness enhancement, and every other heavy-ion observable.

---

## WEEK 9 *Symmetry Groups, SU(3), & the Quark Model*

**Goal:** Understand the group theory underlying the Standard Model—especially SU(3) flavor and SU(3) color—and how quarks are classified.

### Reading

- **Mann** Ch. 3 (Symmetries: groups, Lie groups, algebras, SO(3)), Ch. 5 (Particle classification), Ch. 15–16 (From nuclei to quarks; the quark model)
- **Griffiths** Ch. 1 § 1.8–1.10 (Quark model review — skim if needed), Ch. 5 § 5.5–5.6 (SU(3) flavor, the Eightfold Way, tensor products)
- **L&B** Ch. 9 (Quantum mechanical transformations, representations)

### Key Concepts to Nail Down

- Lie groups and Lie algebras; generators, structure constants
- SU(2) isospin as warm-up; SU(3) flavor and the Eightfold Way
- Tensor product decompositions: $\mathbf{3} \otimes \mathbf{3} = \mathbf{8} \oplus \mathbf{1}$; $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1}$
- Color $SU(3)_c$ vs. flavor $SU(3)_f$: same math, different physics
- Color confinement as a statement: only color-singlet states are observed

**You've done tensor product decompositions before.** This week is about making sure you can distinguish the *flavor* SU(3) (classification of hadrons) from the *color* SU(3) (gauge symmetry of QCD), since both use the same group-theory machinery but play completely different roles.

## WEEK 10 *Non-Abelian Gauge Theory & the QCD Lagrangian*

**Goal:** Generalize the gauge principle from U(1) to SU(3). Derive the QCD Lagrangian and understand why gluons self-interact.

### Reading

- **Griffiths** § 10.3–10.5 (Local gauge invariance, Yang–Mills theory, chromodynamics)
- **L&B** Ch. 46 (Non-abelian gauge theory: Yang–Mills, dynamics of $W_\mu$, symmetry breaking)
- **Mann** Ch. 18.4 (QCD), Ch. 15.5 (Color)
- **P&S** § 15.1–15.2, § 16.1 (Non-abelian gauge invariance; the Yang–Mills Lagrangian) — *reference for full derivation*

### Key Concepts to Nail Down

- From global to local SU(3): $\psi \to U(x)\psi$ requires 8 gauge fields $A_\mu^a$ (gluons)

- Covariant derivative $D_\mu = \partial_\mu - ig_s T^a A_\mu^a$; generators $T^a = \lambda^a/2$
- Field strength tensor: $F_{\mu\nu}^a$ includes a non-abelian term → gluon self-coupling
- The QCD Lagrangian: quark kinetic + gluon kinetic + quark-gluon vertex + 3-gluon + 4-gluon vertices
- Contrast with QED: photons don't self-interact; gluons do → this changes *everything*

## WEEK 11 *QCD Phenomenology: Color Factors, Asymptotic Freedom*

**Goal:** Compute color factors for QCD processes, understand asymptotic freedom and confinement, and connect to $e^+e^- \to$ hadrons and R ratio.

### Reading

- **Griffiths** Ch. 8 (Electrodynamics and chromodynamics of quarks: hadron production in $e^+e^-$, color factors, pair annihilation in QCD, asymptotic freedom)
- **Mann** Ch. 17 (Testing the quark model: hadron production, deep inelastic scattering, structure functions), Ch. 18 (Heavy quarks and QCD)
- **L&B** Ch. 34.4 (Asymptotic freedom via the renormalization group)
- **Bartke** Ch. 2 (QCD and the phase transition: running $\alpha_s$, asymptotic freedom, deconfinement, the QCD phase diagram)

### Key Concepts to Nail Down

- Color factors $C_F = 4/3$, $C_A = 3$, $T_F = 1/2$ for SU(3); how to compute them from traces
- The β-function of QCD: $\beta_0 = 11 - 2n_f/3 > 0$ for $n_f \leq 16$ → asymptotic freedom
- Running coupling $\alpha_s(Q^2)$: strong at low Q, weak at high Q — this is why perturbative QCD works for hard processes in heavy-ion collisions
- $R = \sigma(e^+e^- \to \text{hadrons})/\sigma(e^+e^- \to \mu^+\mu^-) = N_c \Sigma e_q^2$ as evidence for color
- Confinement: $\alpha_s$ blows up at $\Lambda_{QCD} \sim 200$ MeV; perturbation theory breaks down

## WEEK 12 *Weak Interactions & Electroweak Unification ★*

**Goal:** Complete the Standard Model picture with the weak force, CKM mixing, and the Higgs mechanism. Lighter week—focus on the big ideas.

### Reading

- **Griffiths** Ch. 9 (Weak interactions: charged/neutral currents, CKM matrix), Ch. 10.6–10.9 (Feynman rules for weak interactions, mass term, spontaneous symmetry breaking, Higgs mechanism)
- **L&B** Ch. 47 (Weinberg–Salam model: Higgs field, symmetry breaking, gauge boson masses, electron mass)
- **L&B** Ch. 26 (Broken symmetry: Goldstone modes, Higgs mechanism in gauge theories)
- **Mann** Ch. 19–22 (Weak interactions, quark mixing, the Higgs, Standard Model synthesis) — *skim for big picture*

### Key Concepts to Nail Down

- Chirality: weak interaction couples only to left-handed fermions
- $SU(2)_L \times U(1)_Y \rightarrow U(1)_{EM}$ via the Higgs mechanism
- Spontaneous symmetry breaking $\rightarrow$ Goldstone theorem $\rightarrow$ Goldstone bosons "eaten" by $W^{\pm}$, Z
- CKM matrix and flavor mixing; CP violation (relevant to matter–antimatter asymmetry)
- The full Standard Model Lagrangian as a product of gauge principles + symmetry breaking

**Checkpoint — connect to experiment:** Electroweak physics is not the focus of heavy-ion research, but the Higgs mechanism and spontaneous symmetry breaking are essential for understanding *chiral* symmetry breaking in QCD—the mechanism that gives constituent quarks their effective mass and whose restoration is one of the key signatures of QGP formation.

# Phase IV: Bridges to Heavy-Ion Physics

The final two weeks connect everything you've built to the physics you actually measure. This phase is about seeing how QFT's structures—symmetry breaking, the QCD vacuum, and thermal field theory—give rise to the quark-gluon plasma and its experimental signatures.

---

## WEEK 13 *Chiral Symmetry, the QCD Vacuum & the Phase Diagram*

**Goal:** Understand chiral symmetry breaking in QCD, the quark condensate, and the theoretical basis for the QCD phase diagram.

### Reading

- **Bartke** Ch. 2 (QCD and the phase transition: chiral symmetry, order of the transition, lattice QCD predictions, the phase diagram)
- **L&B** Ch. 26 (Broken symmetry: Landau theory, breaking continuous symmetry, Goldstone modes) — *re-read with QCD in mind*
- **P&S** § 19.3 (Chiral symmetry breaking in QCD; the σ model) — *optional deep dive*
- **Bartke** Ch. 21 (More about QGP: chiral condensate, color glass condensate, disoriented chiral condensate)

### Key Concepts to Nail Down

- Chiral symmetry $SU(2)_L \times SU(2)_R$ in the massless quark limit
- Spontaneous chiral symmetry breaking: $\langle q\bar{q} \rangle \neq 0 \rightarrow$ pions as pseudo-Goldstone bosons
- Why $m_\pi \ll m_\rho$: pions are light *because* chiral symmetry is only approximately broken
- The QCD phase diagram: temperature vs. baryo-chemical potential $\mu_B$; crossover vs. first-order transition; critical endpoint
- Lattice QCD as the non-perturbative tool: discretize spacetime, compute path integrals numerically
- Deconfinement transition: at $T_c \sim 155$ MeV, chiral symmetry is (approximately) restored; quarks and gluons become the relevant degrees of freedom

## WEEK 14 *QGP Signatures & Connecting QFT to Your Data*

**Goal:** Map the QFT concepts you've learned onto the experimental observables of the quark-gluon plasma that you measure with sPHENIX and STAR.

### Reading

- **Bartke** Ch. 12 (Strangeness enhancement), Ch. 15 (Collective flow), Ch. 16 (Charmonium suppression: J/ψ as a QGP thermometer)
- **Bartke** Ch. 17–18 (Di-lepton mass spectrum, direct photons), Ch. 19–20 (High-$p_T$ suppression, jet quenching)
- **Bartke** Ch. 14 (HBT/femtoscopy: space-time extent of the source)

• **Bartke** Ch. 13 (Light nuclei, antinuclei, hypernuclei — relevant to your antihypertriton work with STAR)

## Key Concepts to Map

• **Asymptotic freedom** → why hard probes (jets, heavy quarks) are calculable with pQCD

• **Color confinement** → why we see hadrons, not quarks; why deconfinement is extraordinary

• **Chiral symmetry restoration** → modifications to ρ/ω spectral functions (dilepton excess)

• **Strangeness enhancement** → lower threshold for s■s production in QGP vs. hadronic matter

• **J/ψ suppression** → Debye screening of the Q■Q potential in the deconfined medium

• **Jet quenching / $R_{AA}$** → energy loss of hard partons traversing the QGP; directly connected to pQCD cross sections and your $\pi^0$ measurements

• **Collective flow** → hydrodynamic behavior of the QGP; nearly perfect liquid with η/s ~ 1/4π

**Final synthesis:** By this point you should be able to trace a clear line from the QFT Lagrangian → QCD → asymptotic freedom → deconfinement → QGP signatures → the $\pi^0$ spectra and $R_{AA}$ you extract from sPHENIX calorimeter data. That's the whole story from first principles to your terminal.

# Supplementary: Path Integrals & Thermal Field Theory

These topics are important for a deeper understanding of QGP theory but are optional for a first pass through this syllabus. Return to them after completing the 14-week core, or interleave them with Phase III/IV if you have bandwidth.

---

## Path Integrals (2–3 weeks if pursued)

- **L&B** Ch. 23–25 (Path integrals, field integrals, statistical field theory / Wick rotation)
- **P&S** § 9.1–9.3 (Functional methods: path integrals for scalar fields, functional quantization of QED)

Path integrals are the natural language of lattice QCD and thermal field theory. The key insight is that the partition function $Z = \mathrm{Tr}(e^{-\beta H})$ in statistical mechanics maps onto the Euclidean path integral via a Wick rotation $t \to -i\tau$, with $\tau$ compactified on $[0, \beta]$ where $\beta = 1/T$. This is how lattice QCD computes thermodynamic quantities and the QCD phase transition.

## Thermal Field Theory & Finite-Temperature QCD

- **L&B** Ch. 25 (Statistical field theory: Wick rotation, partition function, perturbation theory at finite temperature)
- **Bartke** Ch. 2 (Phase transition: lattice results, critical temperature, equation of state)
- Kapusta & Gale, *Finite-Temperature Field Theory* — the standard reference if you want to go further

Thermal field theory extends QFT to systems at finite temperature and density—exactly the conditions created in heavy-ion collisions. The Matsubara formalism (imaginary-time path integral with periodic/antiperiodic boundary conditions for bosons/fermions) is the backbone of all theoretical predictions for QGP thermodynamics, transport coefficients, and spectral functions.

# Study Strategy

---

**Derive, don't just read.** For every major result (Klein–Gordon mode expansion, Dirac spinor completeness, QED vertex factor, QCD β-function), close the book and re-derive it on paper. This is especially important for the trace technology in Ch. 7–8 of Griffiths—you need to be able to do these in your sleep.

**Use L&B for intuition, Griffiths for computation, P&S for depth.** L&B gives you the physical picture and the "why." Griffiths forces you to actually compute cross sections and decay rates. P&S is the full formal treatment—consult it when L&B or Griffiths hand-wave over something.

**Connect every concept to an observable.** After each week, ask: how does this show up in my data? How would I explain this to someone at a STAR/sPHENIX meeting? The Bartke book is your anchor for these connections.

**Work the exercises.** Reading QFT without doing problems is like reading about swimming. Prioritize exercises that involve computing a cross section, a decay rate, or a color factor—these are the skills with

the most direct payoff.

**Don’t get stuck on rigor.** You’re an experimentalist. You need to understand what the theory says and how to use it, not prove every theorem from scratch. If a derivation is taking more than 2 hours, move on and come back to it.

**Keep a formula sheet.** Build a running document of key results: propagators, Feynman rules for QED and QCD, trace identities, color factors, Mandelstam variables. This becomes an invaluable reference for reading papers.

## Recommended Reading Order Within Each Week

1. Start with L&B for the physical motivation and big picture. 2. Work through Griffiths (or Mann) for the computational details and examples. 3. Do the exercises. 4. Consult P&S only where you need the full, rigorous treatment. 5. End the week by reading the relevant Bartke section to see where the physics meets your detector.

---

*Good luck, Parker. By week 14 you’ll be able to trace the line from $\blacksquare_{QCD}$ all the way to your $\pi^0$ invariant mass plots.*

**— Friday**